Wolstenholme numbers and a gcd conjecture of Eldar and Ordowski (OEIS A007406): infinite families, formalized in Lean 4

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Abstract

Let $a(n)$ be the numerator of $H_n^{(2)} = \sum_{k=1}^{n} k^{-2}$ (OEIS A007406 [1]). Eldar and Ordowski conjectured in 2019 that $\gcd(n, a(n-1))$ equals n for primes $n > 3$ and equals 1 for composites $n > 3$. **The full conjecture remains open.** We prove a leading-digit criterion: if $q \le m$ is prime and q does not divide $a(U)$, where U is the leading base-q digit of m, then $q \nmid a(m)$. As consequences we show that $\gcd(6, a(m)) = 1$ for all $m \ge 1$, and that the conjecture holds for every $n > 3$ that is prime (via Wolstenholme's theorem, which we formalize), 3-smooth, of the form $2q^k$ with q an odd prime, and more generally cq^k with $c \ge 2$ 3-smooth and $q > c$ a prime not dividing $a(c-1)$. All results are formalized in Lean 4 with Mathlib against the definitions in `formal-conjectures` [2], with no `sorry` and only the standard axioms.

1 Introduction

OEIS A007406 [1] contains the comment: *"Conjecture: for $n > 3$, $\gcd(n, a(n-1)) = \mathrm{A089026}(n)$. Checked up to $n = 10^5$."* (A. Eldar and T. Ordowski, Jul. 28, 2019). Here $\mathrm{A089026}(n) = n$ if n is prime and 1 otherwise. J. Sondow added that the prime case follows from Wolstenholme's theorem and that *"it remains to show that $\gcd(n, a(n-1)) = 1$ if $n > 3$ is composite."* The conjecture is formalized as `OeisA7406.conjecture` in `FormalConjectures/OEIS/7406.lean` of `formal-conjectures` [2], where it is tagged `research open`.

2 Related work

Our key lemma is not new in substance. It is the order-2 analogue of the classical leading-digit argument for the sets $J(p) = \{n : p \mid \text{numerator of } H_n\}$, studied by Eswarathasan and Levine [3], Boyd [4], Sanna [6] and Wu and Chen [7]; Kamano [5] treats 3-adic valuations of generalized harmonic numbers. Wolstenholme's theorem and its refinements are surveyed by Meštrović [8]. As of October 12, 2026, the OEIS still lists the conjecture as open, and we found no work applying these tools to it. Our contribution is the reduction to a leading-digit criterion, the explicit infinite families, and machine-checked proofs stated against the upstream definitions (`formal-conjectures` commit `34ee0e3`). The formalization uses Lean 4 [10] and Mathlib [9].

3 Statement

Theorem 1. *The equality* $\gcd(n, a(n-1)) = \mathrm{A089026}(n)$ *holds for every* $n > 3$ *that satisfies one of the following:*

1. *n is prime;*
2. *$n = 2^i 3^j$;*
3. *$n = cq^k$ with $c = 2^i 3^j \geq 2$, $k \geq 1$, and $q > c$ a prime with $q \nmid a(c-1)$. This includes $n = 2q^k$ for every odd prime q, $n = 3q$ for $q \geq 7$, and $n = 4q$ for $q \geq 5$, $q \neq 7$.*

Cases not covered include $n = q^k$ with $k \geq 2$, and general $n = pq$ in which neither cofactor is 3-smooth.

4 Proof overview

4.1 The leading-digit lemma

Throughout, q is prime and v_q is the q-adic valuation on $\mathbb{Q}$.

Lemma 2. *Let $m \geq 1$, $s = \lfloor \log_q m \rfloor$ and $U = \lfloor m/q^s \rfloor$, so $1 \leq U < q$. If $q \nmid a(U)$ then $v_q(H_m^{(2)}) = -2s$.*

Proof. Since $U < q$, every term u^{-2}, $1 \leq u \leq U$, is a q-adic unit, so $v_q(H_U^{(2)}) \geq 0$; as $q \nmid a(U)$ and the denominator of $H_U^{(2)}$ is coprime to q, $v_q(H_U^{(2)}) = 0$. Split $\{1, \ldots, m\}$ into $S_1 = \{k : q^s \mid k\} = \{q^s u : 1 \leq u \leq U\}$ and S_0. Then $\sum_{S_1} k^{-2} = q^{-2s} H_U^{(2)}$ has valuation $-2s$. For $k \in S_0$ we have $v_q(k) \leq s-1$, so $v_q(k^{-2}) \geq -2s+2$; since all terms are positive, the ultrametric inequality gives $v_q(\sum_{S_0} k^{-2}) \geq -2s+1$. Hence $v_q(H_m^{(2)}) = -2s$. □

Theorem 3 (criterion)**.** *If $q \leq m$ and $q \nmid a(\lfloor m/q^{\lfloor \log_q m \rfloor} \rfloor)$, then $q \nmid a(m)$. Consequently, if $n > 3$ is composite and for every prime $q \mid n$, q does not divide a of the leading base-q digit of $n-1$, then* $\gcd(n, a(n-1)) = 1$.

Proof. Here $s \geq 1$, so $v_q(H_m^{(2)}) < 0$; if $q \mid a(m)$ then $v_q(H_m^{(2)}) \geq 1$. For the second claim, a prime divisor q of composite n satisfies $q < n$, so $q \leq n-1$. □

4.2 Consequences

Theorem 4. *For all $m \geq 1$,* $\gcd(6, a(m)) = 1$.

Proof. For $q = 2$ the leading digit is 1 and $a(1) = 1$; for $q = 3$ it is 1 or 2, and $a(1) = 1$, $a(2) = 5$. Small $m < q$ are checked directly. □

Corollary 5. *The conjecture holds for every $n = 2^i 3^j > 3$.*

Theorem 6. *Let $c = 2^i 3^j \geq 2$ and let $q > c$ be prime with $q \nmid a(c-1)$. Then* $\gcd(cq, a(cq-1)) = 1$, *that is, the conjecture holds at $n = cq$.*

Proof. Primes $2, 3$ are handled by Theorem 4. For q: $q \leq cq - 1 < q^2$, so $s = 1$ and the leading digit is $\lfloor (cq-1)/q \rfloor = c-1$. □

Corollary 7. *The conjecture holds for $n = 2q$ (all primes $q \geq 3$), $n = 3q$ (primes $q \geq 7$), and $n = 4q$ (primes $q \geq 5$, $q \neq 7$), using $a(1) = 1$, $a(2) = 5$, $a(3) = 49$. For each 3-smooth $c \geq 2$ it holds for $n = cq$ for all primes $q > \max(c, a(c-1))$.*

Theorem 8. *Let $c = 2^i 3^j \geq 2$, $q > c$ prime with $q \nmid a(c-1)$, and $k \geq 1$. Then the conjecture holds at $n = cq^k$. In particular it holds for $n = 2q^k$, for every odd prime q and every $k \geq 1$.*

Proof. For $m = cq^k - 1$ we have $q^k \leq (c-1)q^k \leq m < cq^k < q^{k+1}$, so $\lfloor \log_q m \rfloor = k$ and the leading digit is $c-1$. Primes $2, 3$ are handled by Theorem 4. □

4.3 The prime direction

Theorem 9 (Wolstenholme)**.** *For every prime $p \geq 5$, $p \mid a(p-1)$. Hence* $\gcd(p, a(p-1)) = p =$ A089026(p) *for every prime $p > 3$.*

Proof. Let $F = (p-1)!$ and $N = \sum_{k=1}^{p-1}(F/k)^2$, so that $H_{p-1}^{(2)}F^2 = N$. In $\mathbb{F}_p$, Wilson gives $k \cdot (F/k) = -1$, so $(F/k)^2 = (F/k)^2 k^{p-1} = k^{p-3}$. Therefore $N \equiv \sum_{x \in \mathbb{F}_p} x^{p-3} = 0$, since $0 < p-3 < p-1$. As $p \nmid F^2$, p divides the numerator. □

Corollary 10. *The full conjecture* $\gcd(n, a(n-1)) =$ A089026(n) *holds for every $n > 3$ that is prime, 3-smooth, or of the form $2q^k$ (q an odd prime, $k \geq 1$), and for $n = cq^k$ as in Theorem 8.*

Remark 11 (Coverage and limits). A short computation (`coverage.py`) shows the criterion of Theorem 3 settles 76,100 of the 90,407 composites $4 \leq n \leq 10^5$ (about 84%). It cannot settle every case: $q \mid a(q-1)$ (Wolstenholme) and $q \mid a((q-1)/2)$ for all $q \geq 5$, so for instance $n = q^2$ needs a second-order analysis. There $H_{q^2-1}^{(2)} = \sum_{q \nmid k < q^2} k^{-2} + q^{-2}H_{q-1}^{(2)}$, and the first sum is $\equiv 0 \pmod{q^2}$. One finds that $q \mid a(q^2-1)$ would force $v_q(H_{q-1}^{(2)}) \geq 3$, which by Glaisher's congruence $H_{q-1}^{(2)} \equiv \frac{2}{3}qB_{q-3} \pmod{q^2}$ requires $q \mid B_{q-3}$ (a Wolstenholme prime, e.g. 16843) together with a further congruence. We do not formalize this.

5 Formalization

All declarations are in namespace `Wolst`. The definitions `a` and `A89026` in `Basic.lean` are copied verbatim from `formal-conjectures`, and every main theorem concludes exactly `Nat.gcd n (a (n - 1)) = A89026 n`.

Lean name (file)	Statement
`padicValRat_H`, `not_dvd_a` (`Basic.lean`)	Lemma 2; first part of Theorem 3
`conjecture_of_leading_digits` (`Families.lean`)	Theorem 3
`coprime_six_a`, `conjecture_smooth`	Theorem 4 and its corollary
`conjecture_smooth_mul_prime`	Theorem 6
`conjecture_two/three/four_mul_prime`, `conjecture_smooth_mul_large_prime`	corollaries of Theorem 6
`conjecture_smooth_mul_prime_pow`, `conjecture_two_mul_prime_pow` (`Prime.lean`)	Theorem 8
`prime_dvd_a`, `conjecture_prime`	Theorem 9
`conjecture_on_families`	cases (1)–(2) and $2q^k$ of Theorem 1

Table 1: Theorems and their Lean names.

6 Reproducibility

- Source code: `https://github.com/het4rk/eldar-ordowski-a007406`. The Lean project is in `Wolst/`.

- Toolchain: `leanprover/lean4:v4.35.0-rc4`. Mathlib: tag `v4.35.0-rc4`, commit `1f414401f69059aa7eead47b53ee40bd38455eeb`, pinned in `lake-manifest.json`.
- Build:

```
cd Wolst
lake exe cache get
lake build
lake env lean Axioms.lean
```

- Axioms: each of the 15 declarations in `Axioms.lean` uses only the axioms `propext`, `Classical.choice` and `Quot.sound`.
- Attestation: `verification/` contains the fresh-clone build log, the axiom output and `VERIFICATION.md`, which records the commit, the toolchain and SHA-256 hashes of the Lean sources. CI (`https://github.com/het4rk/eldar-ordowski-a007406/actions/workflows/lean.yml`) rebuilds the project on every push. It rejects `sorry` and non-whitelisted axioms, and uploads the build log and axiom report as artifacts.

7 Limitations

The full conjecture remains open. The criterion can never settle $n = q^2$, because $q \mid a(q-1)$; a second-order analysis would be needed (Remark 11), and it is not formalized. The coverage figure in that remark was computed by `paper/coverage.py`. It is not a formal result.

References

[1] OEIS Foundation Inc., *A007406: Wolstenholme numbers*, `https://oeis.org/A007406`.

[2] Google DeepMind, *formal-conjectures*, `https://github.com/google-deepmind/formal-conjectures`, file `FormalConjectures/OEIS/7406.lean`.

[3] A. Eswarathasan, E. Levine, p-integral harmonic sums, *Discrete Math.* **91** (1991), 249–257.

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[6] C. Sanna, On the p-adic valuation of harmonic numbers, *J. Number Theory* **166** (2016), 41–46.

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[8] R. Meštrović, Wolstenholme's theorem: its generalizations and extensions in the last hundred and fifty years, arXiv:1111.3057.

[9] The mathlib Community, *The Lean mathematical library*, in Proc. CPP 2020, ACM, 367–381.

[10] L. de Moura, S. Ullrich, *The Lean 4 theorem prover and programming language*, in CADE-28, LNCS 12699, Springer, 2021, 625–635.